

```
1 clc;
2 clear;
3 close all;
4
5 % =====
6 % SNR values (dB)
7 % =====
8 SNR = 0:5:30;
9
10 % =====
11 % 5G Throughput
12 % =====
13 Max_TP = 100;
14
15 % Throughput calculation
16 Throughput = Max_TP*(1-exp(-SNR/10));
17
18 disp('SNR(dB) Throughput(Mbps)')
19 disp([SNR' Throughput'])
20
21 % =====
22 % BER and Reliability
23 % =====
24 BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];
25
26 % Reliability (%)
27 Reliability = (1-BER)*100;
28
29 disp('SNR(dB) BER Reliability(%)')
30 disp([SNR' BER' Reliability'])
31
32 % =====
33 % Modulation Schemes
34 % =====
35 Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};
36
37 % Required SNR (dB)
38 Mod_SNR = [5 10 18 25];
39
40 % Spectral Efficiency (bits/s/Hz)
41 Efficiency = [2 4 6 8];
42
43 disp('Required SNR (dB) and Spectral Efficiency')
44
45 disp(table(Modulation,Mod_SNR,Efficiency', ...
46
47
```

Figure 1

